

Manuel Fernández López

Postdoctoral Researcher, Lattice Field Theory Group

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Summary

Condensed matter theorist specializing in topologically ordered, strongly correlated systems and their applications to quantum information science.

Education

PhD	Autonomous University of Madrid (UAM), Physics - Thesis: "Fractionalized topological phases in strongly correlated materials with spin-orbit coupling" - Supervisor: Prof. Jaime Merino - Defense: Oct 2024 — Cum laude	Madrid, Spain 2019 – 2024
MSc	Autonomous University of Madrid (UAM), Condensed Matter Physics and Biological Systems Thesis: "Spin-orbit coupling and Coulomb interaction effects in decorated honeycomb lattices"	Madrid, Spain 2017 – 2018
BSc	Autonomous University of Madrid (UAM), Physics - Theoretical final project: "Bosonic string theory for the derivation of the graviton and compactification of 25+1 dimensions." - Experimental final project: "Data analysis for the determination of the Z0 boson mass using LHC data."	Madrid, Spain 2013 – 2017

Experience

Institute of Physics, National Yang Ming Chiao Tung University, Postdoctoral Researcher, Lattice Field Theory Group - Member of the Lattice Field Theory Group at NYCU, translating computational techniques between condensed matter physics and non-perturbative QCD. - Applying parton-based and exact diagonalization methods developed for strongly correlated, topologically ordered systems to open questions in hadron structure. - Building complementary expertise in tensor network and DMRG approaches to lattice gauge theories.	Hsinchu, Taiwan Sept 2026 – present 1 month
Autonomous University of Madrid (UAM), Associate Researcher - Developed theoretical approaches to compute Green's function zeros in topological Mott insulators; application to the Kitaev-Hubbard model (preprint in final stage). - Co-supervised a PhD project on unconventional superconductivity in nickelate heterostructures using semi-analytical and numerical many-body techniques. - Proposed novel topological strongly correlated systems to apply AI-driven methodologies developed within the group. - Prepared a referee report for Physical Review Letters; regularly handled editorial/referee correspondence as corresponding/co-author. - Acted as corresponding author on the first publication of a PhD student, structuring the manuscript and overseeing the full reviewer-editor correspondence.	Madrid, Spain Nov 2024 – Sept 2026 1 year 11 months

Autonomous University of Madrid (UAM), Researcher Formation Fellow (FPI)

- Implemented a finite-T slave-rotor mean-field framework for topological Mott transitions; benchmarked local limits against DMFT.
- Identified flat-band topological superconductivity and localized Majorana edge modes in metal-organic frameworks (Phys. Rev. B 106, 235129, 2022).
- Performed slab Green's-function calculations revealing spinon Fermi arcs, a novel type of topological surface state (Phys. Rev. B 109, 085137, 2024).
- Derived effective Hubbard models from first-principles DFT band structures for strongly correlated-material modeling (Phys. Rev. B 111, 165133, 2025).

Madrid, Spain
Sept 2019 – Oct 2024
5 years 2 months

Autonomous University of Madrid (UAM), Scientific Technician, National Research Project

- Adapted master's thesis results into a peer-reviewed publication and managed correspondence with editors and referees (Phys. Rev. B 100, 075154, 2019).
- Prepared departmental talks and presented results at a national conference.

Madrid, Spain
Mar 2018 – Aug 2019
1 year 6 months

Research Projects

Tensor networks and quantum computing for high-energy physics

NSTC grant numbers (verified against Francis, Lin, Morris & Zhao, arXiv:2606.19221):
112-2112-M-A49-021-M03, 114-2123-M-A49-001-ASP, 114-2811-M-A49-516.

NSTC special-project grant;
PI: Prof. C.-J. David Lin,
Lattice Field Theory Group,
NYCU
2026 – present

New concepts and methods for the comprehension and characterization of topological strongly correlated quantum matter Ref. PID2022-139995NB-I00;
IP: Jorge Bravo Abad

- Computed non-local Green's function in-gap zeros delineating nodal-line, charge-neutral low-energy excitations — a fractionalized protected state (spinon drumhead states).
- Led an individual research line computing Green's function zeros in Kitaev materials such as α -RuCl₃, opening a route to identify topological phases via GF zeros in correlated superconductors.

New phenomena and applications of topological strongly correlated materials in and out of equilibrium Ref. RTI2018-098452-B-I00;
IP: Jaime Merino Troncoso
(FPI PRE2019-087603)

- Supervised a master's student on code adaptation for correlated nodal-line semimetals relevant to (ET)Ag₄(CN)₅ compounds, providing the first theoretical realization of the observed pressure-induced Mott transition.
- Developed a semi-analytical strongly correlated technique extended to finite temperature, revealing that non-local correlations beyond DMFT lead to spinon Fermi arcs, with predicted ARPES signatures on pyrochlore iridates.
- Investigated correlated topology in flat-band decorated honeycomb lattices motivated by metal-organic frameworks (MOFs); exact diagonalization revealed a flat-band quantum spin liquid consistent with experiments.
- Explored flat-band unconventional superconductivity from doping this spin liquid, characterizing localized Majorana edge modes and their potential for topological quantum computing.

New electronic and photonic materials based on topological strongly correlated systems Ref. MAT2015-66128-R; IP: Jaime Merino Troncoso

- Implemented Hartree-Fock calculations in a decorated honeycomb lattice relevant to certain MOFs, identifying topological semimetals.
- Presented findings at national conferences and acted as corresponding author in the peer-review process.

Teaching & Mentoring

Co-supervisor of a Master's and current Ph.D. student

- Mentoring on topological strongly correlated electron models and Green's function zeros analysis, resulting in a joint publication.
- As corresponding author on the student's first paper, guided the full publication pipeline (submission, revision, peer-review dialogue) while enforcing good practices in code/documentation and reproducible workflows.

Autonomous University of Madrid (UAM)
2024 – present

Lecturer & Lab Supervisor, Physics Laboratory course

Environmental Science Degree (2nd year), Autonomous University of Madrid (UAM)
2021 – 2022

Peer-Review Activity

- Referee report prepared for Physical Review Letters (submitted Aug 2025; editorial decision Sep 2025), assessing significance, methodological soundness and clarity of presentation.
- Regularly manage reviewer/editor correspondence for co-authored and first-authored manuscripts, including coordinating responses, revising manuscripts, and ensuring reporting/figures meet journal standards.
- Peer reviewer for Physical Review B, first as co-referee with supervisor, later as an independent referee.
- Experience assessing reproducibility, figure clarity, methodological soundness and experimental relevance in theoretical manuscripts, with a focus on transparent availability of data and custom code.

Presentations

- *From Quantum Anomalous Hall Phases to Topological Metals in Interacting Decorated Honeycomb Lattices* — Oral contribution, INC Young Researchers Meeting, Madrid (2019)
- *Magnetism and Topological Phases in an Interacting Decorated Honeycomb Lattice with Spin-Orbit Coupling* — Oral contribution, INC Young Researchers Meeting, Madrid (2021)
- *Topological Phases Induced by Interactions and Spin-Orbit Coupling in Decorated Honeycomb Lattices* — Oral contribution, APS March Meeting, online (2021)
- *Bad Topological Semimetals in Layered Honeycomb Compounds* — Oral contribution, APS March Meeting, Chicago, IL, USA (2022)
- *Bad Topological Semimetals in Layered Honeycomb Compounds* — Invited talk, School of Mathematics and Physics, University of Queensland, Australia (2022)
- *Emergence of Spinon Fermi Arcs in the Weyl-Mott Metal-Insulator Transition* — Oral contribution, INC Young Researchers Meeting, Madrid (2024)
- *Bad Weyl Semimetals and Spinon Fermi Arcs in a Model for Pyrochlore Iridates* — Oral contribution, IOP CMQM 2024, St. Andrews, Scotland (2024)

Skills

Semianalytical / Numerical: Parton mean-field theories, exact diagonalization, solid working knowledge of (cluster) DMFT and QMC methods, topological invariants calculation in the absence of Landau quasiparticles

Programming & Software: LaTeX, Fortran, Matlab (high proficiency); Adobe Suite, Wolfram Mathematica (advanced); Python (beginner); Windows, macOS, Linux

Communication: Manuscript evaluation, referee correspondence, figure/table preparation

Languages: English (fluent), Spanish (native)

Awards

FPI Research Fellowship

Autonomous University of Madrid (UAM)
2019 – 2024